

Measuring and Mitigating Alignment Risks of Language Models

Roma Patel

My research studies how language model systems might create and amplify long-term risks through interactions with people, and develops methods to measure and mitigate these socio-affective risks. Language model products are increasingly—and sometimes unknowingly—integrated into people’s lives, making the need to measure and mitigate unsafe behaviors increasingly more urgent. Although humans are perceptive enough to detect subtle biases and forms of manipulation, there is increasing evidence of humans being polarized, manipulated or deceived by language model agents. I focus in particular on failures that emerge in contexts where human preferences, values, and social behavior conflict with alignment objects, and lead to behaviors such as sycophancy, manipulation or the suppression of minority perspectives. This needs to be done in principled academic ways—in collaboration with, but without industry influence and competing incentives. Separately, existing post-training and alignment paradigms incentivize preference agreement with brittle behavioral guardrails that time and again fail under adversarial pressure. My work so far has focused on three aspects of the problem.

- How can we detect and understand occurrences of safety issues in human language model conversations? [TBR: gpt-bias, values]
- How can we improve RLHF/alignment pipelines to make them better? [TBR: sycophancy, sct-rewards, game theory]
- How can we create systems to attack and investigate models? [TBR: leagues, probing, sycophancy]

The first direction of my research asks how language models influence behaviors of humans outside of the immediate content of an individual response. Most safety evaluations and research focus on the immediate context of the chat interaction, and the influence on the human participant, but not longer-term influences. My work has developed evaluations to detect subtle behavioral effects of language-model outputs over long periods of time. This requires a controlled study setting of repeated interactions over multiple days or weeks of interactions. When coupled with real behavioral outcomes of participants, this data allows us to understand the effect of language interactions and continued exposure to language models

Working at the frontier of deploying language model products has exposed me to a class of alignment problems that become visible only when models reach a sufficient bar of capability, are widely used, and can be projected to have continued longitudinal risks. We identify subtle ways in which language model generations can influence human behaviors, both in short interactions and along very long time scales [gpt-bias]. We detect and control for sycophantic behaviors of models—situations where preference optimization pipelines reward agreeableness and flattery—and how this can pose dangers to humans [control-syc]. We update the reinforcement learning reward formulation using social choice theory aggregation methods to allow for better aggregation of different rewards [sct-rlhf]. We show how intransitivity in human preferences can be better handled by alignment pipelines that handle game-theoretic objectives, and show how this can recover and

handle erasure of minority preferences better. [intransitivity]. We create self-improving adversarial red-teaming systems to uncover dangerous capabilities of models [leagues]. We create datasets for evaluating differing safety perspectives, build models that can handle this better, and also work towards longitudinal measurements of data. [dices, pluralis, long-term]. Building off of this work, my future research will work on answering the following:

- How can alignment objectives be grounded in frameworks that allow for fairer representation of conflicting values and stakeholder preferences?
- How can we design self-improving multi-agent language systems that can surface emerging failure models and evolving risks in models?
- How can we measure and mitigate long-term risks through targeted controlled studies?

References